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**UNIVERSITY OF BATANGAS – LIPA CAMPUS  
COLLEGE OF ENGINEERING AND ARCHITECTURE  
COMPUTER ENGINEERING**

**Seven Segment Display**

**Midterm Project**  
**Logic and Circuit Design**

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## **Objectives:**

At the end of this laboratory exercise, you should be able to:

1. To be able to construct a functional seven-segment display circuit using logic gates.
2. To be able to implement a seven-segment display circuit using Boolean equations derived from a truth table.

## **Equipment and Materials:**

The following equipment and materials are needed in performing this laboratory exercise:

- 1 - DC Power Supply with +5V output capability
- 1 - Digital Multimeter
- 1 - Breadboard
- 1 - 7404 or 74LS04 hex Inverter IC
- 3 - 7408 or 74LS08 Quad 2-input AND gate IC
- 4 – 7432 or 74LS32 Quad 2-input OR gate IC
- 1 – 7 Segment Display
- 4 – Slide Switch (*Optional*)
- Connecting Wires



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## **Discussion:**

Seven-segment display is a fundamental electronic component used to visually represent numerical data in digital systems. It consists of seven individually controlled light-emitting segments arranged in a pattern that allows the formation of digits from 0 to 9. Each segment can be modeled using Boolean variables, where specific combinations correspond to the activation of segments for a particular numeral. The logical relationships governing these combinations can be derived from truth tables and expressed through Boolean functions. These functions are often simplified using Karnaugh maps to minimize the circuit complexity and optimize the display's performance.

The seven-segment display operates in two main configurations: common anode and common cathode. In both types, Boolean simplification plays a crucial role in efficiently controlling which segments are illuminated for each desired output. By applying digital logic design principles, the implementation of the seven-segment display demonstrates the practical integration of Boolean algebra in modern electronic systems. Its design and analysis provide an essential foundation for understanding how abstract logic expressions translate into functional visual displays.



## Procedure:

1. From the truth table below, find the Boolean equations using the KMAP (Karnaugh Map).

| No. | Input |   |   |   | Output |   |   |   |   |   |   |
|-----|-------|---|---|---|--------|---|---|---|---|---|---|
|     | W     | X | Y | Z | a      | b | c | d | e | f | g |
| 0   | 0     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 1   | 0     | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 2   | 0     | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 3   | 0     | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 4   | 0     | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 5   | 0     | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 6   | 0     | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 7   | 0     | 1 | 1 | 1 | 1      | 1 | 1 | 0 | 0 | 0 | 0 |
| 8   | 1     | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 9   | 1     | 0 | 0 | 1 | 1      | 1 | 1 | 1 | 0 | 1 | 1 |
| 10  | 1     | 0 | 1 | 0 | x      | x | x | x | x | x | x |
| 11  | 1     | 0 | 1 | 1 | x      | x | x | x | x | x | x |
| 12  | 1     | 1 | 0 | 0 | x      | x | x | x | x | x | x |
| 13  | 1     | 1 | 0 | 1 | x      | x | x | x | x | x | x |
| 14  | 1     | 1 | 1 | 0 | x      | x | x | x | x | x | x |
| 15  | 1     | 1 | 1 | 1 | x      | x | x | x | x | x | x |



## The Boolean Equations:

|          |                               |
|----------|-------------------------------|
| <b>a</b> | $X'Z' + W + XZ + Y$           |
| <b>b</b> | $Y'Z' + YZ + X'$              |
| <b>c</b> | $Y' + Z + X$                  |
| <b>d</b> | $X'Z' + W + XY'Z + X'Y + YZ'$ |
| <b>e</b> | $X'Z' + YZ'$                  |
| <b>f</b> | $W + XY' + Y'Z' + XZ'$        |
| <b>g</b> | $W + XY' + X'Y + YZ'$         |

## KMAP (Karnaugh map) Solutions:

**a**

| YZ \ WX | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  | 0  | 0  | 1  |
| 01      | 0  | 1  | 1  | 1  |
| 11      | x  | x  | x  | x  |
| 10      | 1  | 1  | x  | x  |

|         |         |         |         |
|---------|---------|---------|---------|
| 1 1 0 0 | 0 0 0 0 | 1 1 0 0 | 0 0 1 1 |
| 1 1 0 1 | 0 0 1 0 | 1 1 0 1 | 0 0 1 0 |
| 1 1 1 1 | 1 0 0 0 | 0 1 0 1 | 0 1 1 1 |
| 1 1 1 0 | 1 0 0 1 | 0 1 1 1 | 0 1 1 0 |
| 1 0 0 0 | 1 0 1 1 | 1 1 0 1 | 1 1 1 1 |
| 1 0 0 1 | 1 0 1 0 | 1 1 1 1 | 1 1 1 0 |
| 1 0 1 1 |         |         | 1 0 1 1 |
| 1 0 1 0 |         |         | 1 0 1 0 |
| X' Z'   | W       | X Z     | Y       |

$$a(W,X,Y,Z) = \sum m ( X'Z' + W + XZ + Y )$$

**b**

| YZ \ WX | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  | 1  | 1  | 1  |
| 01      | 1  | 0  | 1  | 0  |
| 11      | x  | x  | x  | x  |
| 10      | 1  | 1  | x  | x  |

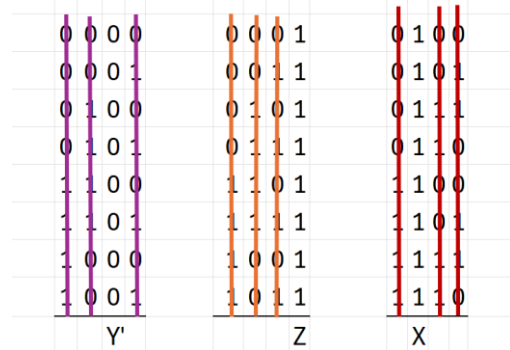
|         |         |         |
|---------|---------|---------|
| 0 0 0 0 | 0 0 1 1 | 0 0 0 0 |
| 0 0 1 0 | 0 1 1 1 | 0 0 0 1 |
| 1 1 0 0 | 1 1 1 1 | 0 0 1 1 |
| 1 1 0 1 | 1 0 1 1 | 0 0 1 0 |
| 1 0 0 0 |         | 1 0 0 0 |
|         |         | 1 0 0 1 |
|         |         | 1 0 1 1 |
|         |         | 1 0 1 0 |
| Y'Z'    | XZ      | X'      |

$$b(W,X,Y,Z) = \sum m ( Y'Z' + YZ + X' )$$



c

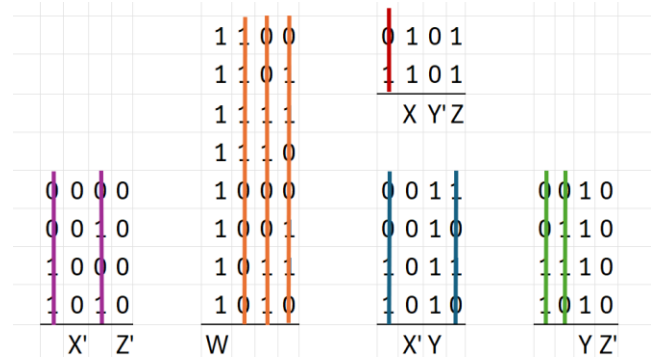
| YZ<br>WX | 00 | 01 | 11 | 10 |
|----------|----|----|----|----|
| 00       | 1  | 1  | 1  | 0  |
| 01       | 1  | 1  | 1  | 1  |
| 11       | x  | x  | x  | x  |
| 10       | 1  | 1  | x  | x  |



$$c(W,X,Y,Z) = \sum m ( Y' + Z + X )$$

d

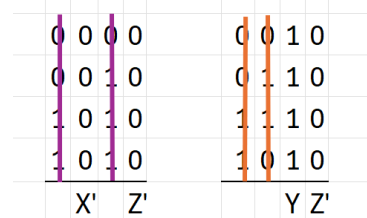
| YZ<br>WX | 00 | 01 | 11 | 10 |
|----------|----|----|----|----|
| 00       | 1  | 0  | 1  | 1  |
| 01       | 0  | 1  | 0  | 1  |
| 11       | x  | x  | x  | x  |
| 10       | 1  | 1  | x  | x  |



$$d(W,X,Y,Z) = \sum m ( X'Z' + W + XY'Z + X'Y + YZ' )$$

e

| YZ<br>WX | 00 | 01 | 11 | 10 |
|----------|----|----|----|----|
| 00       | 1  | 0  | 0  | 1  |
| 01       | 0  | 0  | 0  | 1  |
| 11       | x  | x  | x  | x  |
| 10       | 1  | 0  | x  | x  |

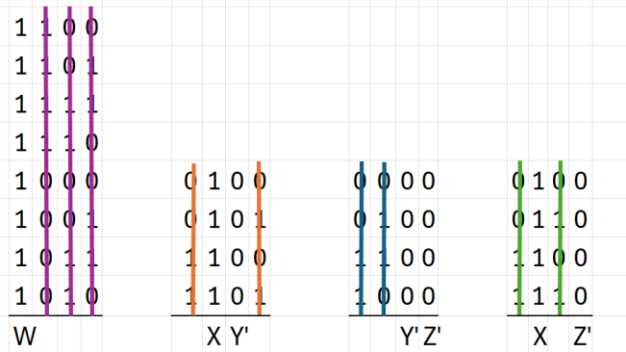


$$e(W,X,Y,Z) = \sum m ( X'Z' + YZ' )$$



f

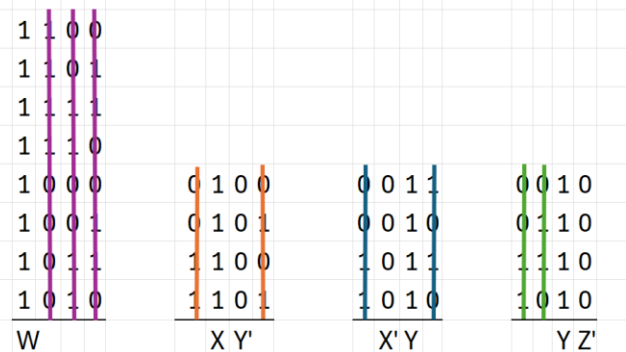
| YZ \ WX | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  | 0  | 0  | 0  |
| 01      | 1  | 1  | 0  | 1  |
| 11      | x  | x  | x  | x  |
| 10      | 1  | 1  | x  | x  |



$$f(W,X,Y,Z) = \sum m ( W + XY' + Y'Z' + XZ' )$$

g

| YZ \ WX | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 0  | 0  | 1  | 1  |
| 01      | 1  | 1  | 0  | 1  |
| 11      | x  | x  | x  | x  |
| 10      | 1  | 1  | x  | x  |

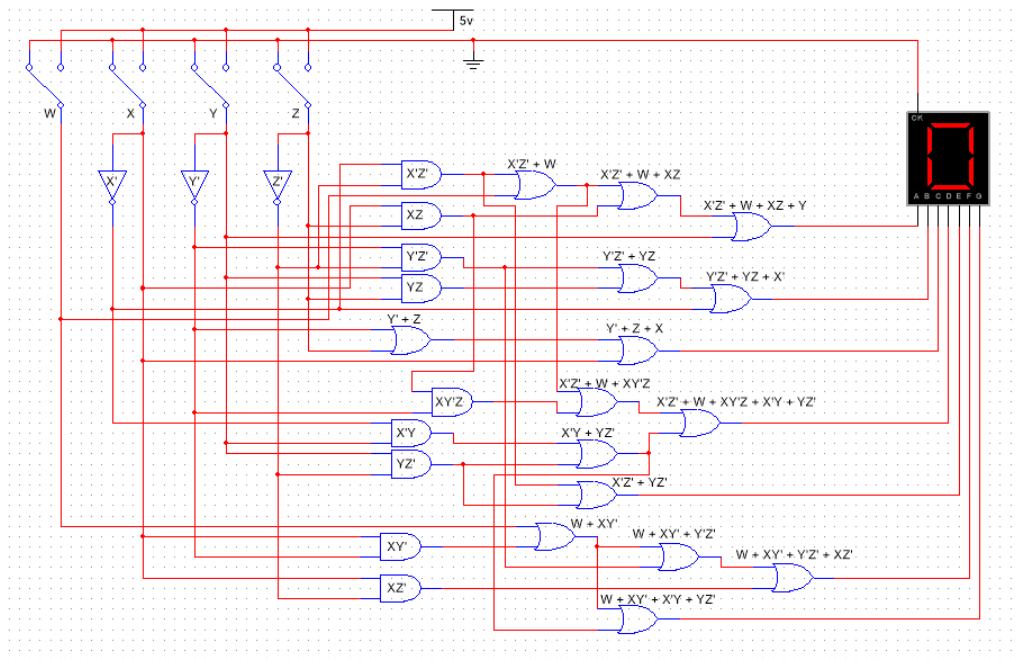


$$g(W,X,Y,Z) = \sum m ( W + XY' + X'Y + YZ' )$$

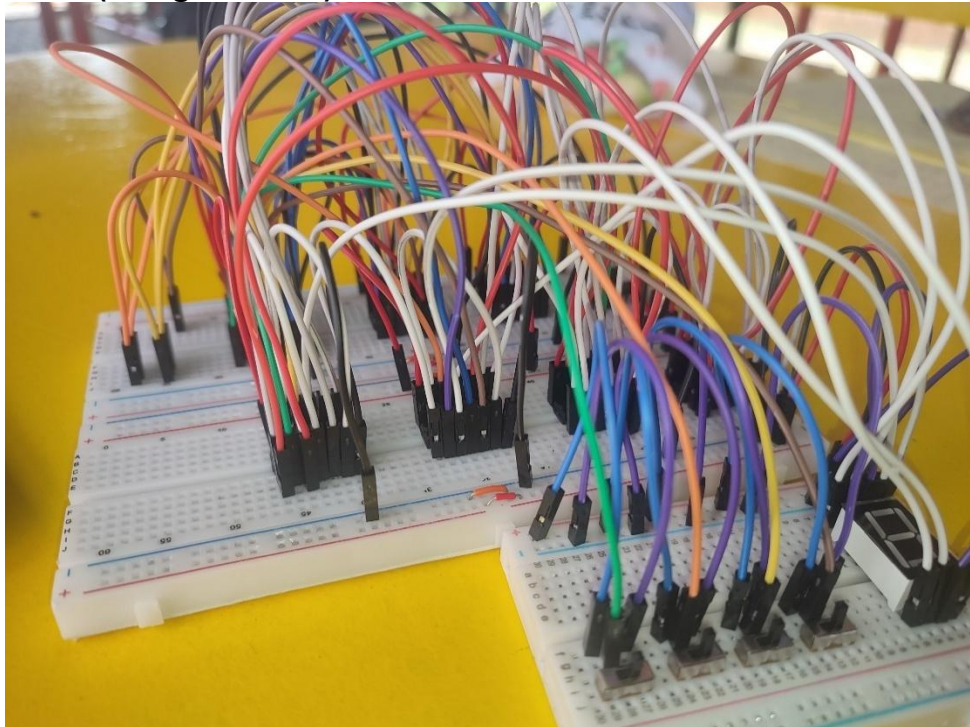


2. From the tables above, apply the Boolean equations to the logic circuit.

**Multisim Diagram (using minterm)**



**Physical Circuit (using minterm)**







## Additional Experiment:

In this part of the experiment, it was observed that expressing the Boolean functions in maxterm form resulted in a more concise and efficient circuit design compared to using minterms. The maxterm, or product-of-sums, approach allowed for the reduction of gate count and interconnections, simplifying the overall circuit without altering its intended output. While minterm expressions explicitly identify all conditions that activate a segment, they often lead to more complex implementations. In contrast, the maxterm formulation focuses on conditions that deactivate outputs, providing opportunities for greater simplification. This finding highlights the practical benefit of analyzing both representations in digital design, as the use of maxterms can lead to shorter, faster, and more cost-effective circuit realizations.

### The Boolean Equations:

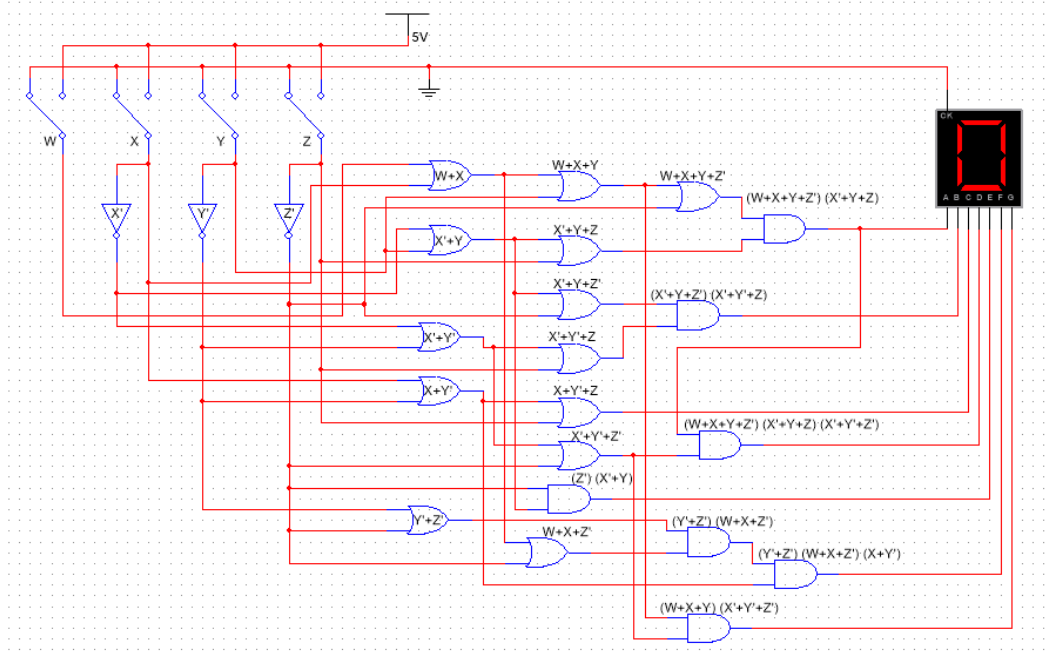
|          |                                  |
|----------|----------------------------------|
| <b>a</b> | $(W+X+Y+Z') (X'+Y+Z)$            |
| <b>b</b> | $(X'+Y+Z') (X'+Y'+Z)$            |
| <b>c</b> | $X + Y' + Z$                     |
| <b>d</b> | $(W+X+Y+Z') (X'+Y+Z) (X'+Y'+Z')$ |
| <b>e</b> | $(Z') (X'+Y)$                    |
| <b>f</b> | $(Y'+Z') (W+X+Z') (X+Y')$        |
| <b>g</b> | $(W+X+Y) (X'+Y'+Z')$             |



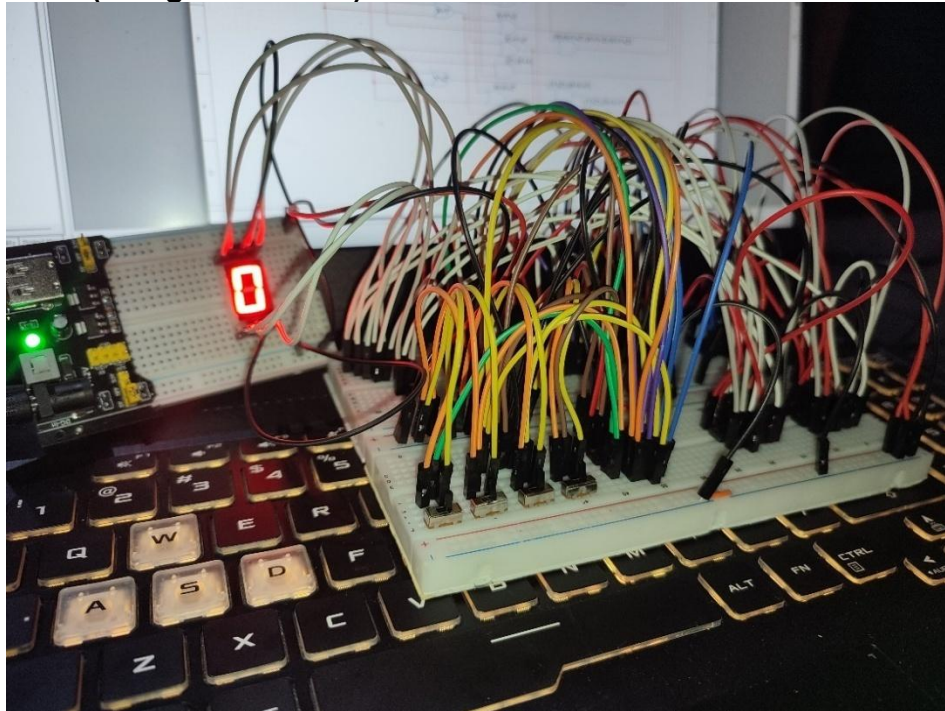
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### Multisim Diagram (using MAXTERM)



### Physical Circuit (using MAXTERM)





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## **Results and Discussion:**

In this laboratory Midterm experiment, we successfully constructed a functional seven-segment display circuit using discrete logic gates, implementing the Boolean equations derived from the truth table with the use of Kmap (Karnaugh Map). The circuit translated the 4-bit binary input (W, X, Y, Z) into the appropriate activation signals (a through g) for the seven-segment display, visibly forming digits 0 through 9. The design we made is based on the minterm (Sum-of-Products) expressions, required a complex arrangement of Inverter (74LS04N), AND (74LS08N), and OR (74LS32N) gate ICs, as shown in the Multisim diagram and physical breadboard. This implementation demonstrated the practical application of Boolean algebra and digital logic design principles to create a visual display system. The circuit successfully integrated abstract logic expressions into a physical, functional circuit. In addition, we also conduct another circuit design with the use of Kmap Maxterm (Products-of-Sum), which we able to make a circuit with fewer Logic gates that still works as the same input from the minterm circuit design.

## **Conclusion:**

In conclusion, we succesfully construct a functional seven-segment display circuit using logic gates and implementing it with Boolean equations derived from a truth table. The experiment confirmed that the minterm expressions accurately represent the conditions needed to activate each segment for the corresponding numerical output. The implementation of the seven-segment display with logic gates serves as an essential foundation for understanding how abstract logic translates into functional electronic components in modern digital systems.



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**Documentation:**

